

JEDEC-STYLE SAMPLE STANDARD

Synthetic DDR6 SDRAM Specification

JESD-SYN-01

Revision 1.0, October 2026

This document is synthetic. It imitates the layout of an industry standard for software testing only and contains invented values.

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Foreword

This synthetic standard was prepared to exercise document conversion tools. It defines a fictional DDR6 SDRAM device with a 16-bit data bus, a data rate of up to 9600 MT/s and a core supply of 1.1 V. All numbers are invented and must not be used for any design.

1 Scope

This document defines the minimum set of requirements for a synthetic x16 DDR6 SDRAM device operating from VDD = 1.1 V and VDDQ = 0.5 V. The device supports densities from 16 Gb to 64 Gb and a refresh interval tREFI of 3.9 μ s at temperatures up to 85 °C.

2 Normative References

The following documents are referenced in this standard:

- JESD-SYN-00, Synthetic Terms and Definitions (invented reference).
- JESD-SYN-02, Synthetic Package Outline Drawings, Revision 2.1.
- ISO-like Test Method TM-17 for $\pm 5\%$ supply tolerance measurement.

3 Terms and Definitions

For the purposes of this standard, the following terms apply.

3.1 tCK(avg)

Average clock period, measured over 200 consecutive cycles. The minimum tCK(avg) is 0.208 ns at 9600 MT/s.

3.2 nCK

Number of clock cycles of CK_t/CK_c, used to express latencies such as RL and WL.

4 Pinout and Signal Description

4.1 Signal Description

Table 1 lists the external signals of the device. Signals with suffix _t and _c form differential pairs; signals with suffix _n are active low.

Table 1 — Signal Description

Symbol	Type	Function
CK_t, CK_c	Input	Differential clock inputs. All address and command inputs are sampled on the crossing of CK_t rising and CK_c falling.
CS_n	Input	Chip select. The command decoder is disabled when CS_n is HIGH.
CA[13:0]	Input	Command/address inputs, sampled at the rising edge of CK_t.
DQ[15:0]	I/O	Bidirectional data bus.
DQS_t, DQS_c	I/O	Data strobe, edge-aligned with read data and centered with write data.
RESET_n	Input	Active-low asynchronous reset; VDD must be stable for 200 μ s before RESET_n is released.
VDD	Supply	Core power supply: 1.1 V $\pm$ 0.033 V.
VDDQ	Supply	I/O power supply: 0.5 V $\pm$ 0.025 V.
VSS	Supply	Ground.

NOTE 1 Unused CA inputs shall be tied to VDDQ through a 240 Ω resistor.

4.2 Ball Assignment

Figure 1 shows the ball assignment as seen from the top of the package. Balls not shown are reserved and shall not be connected.

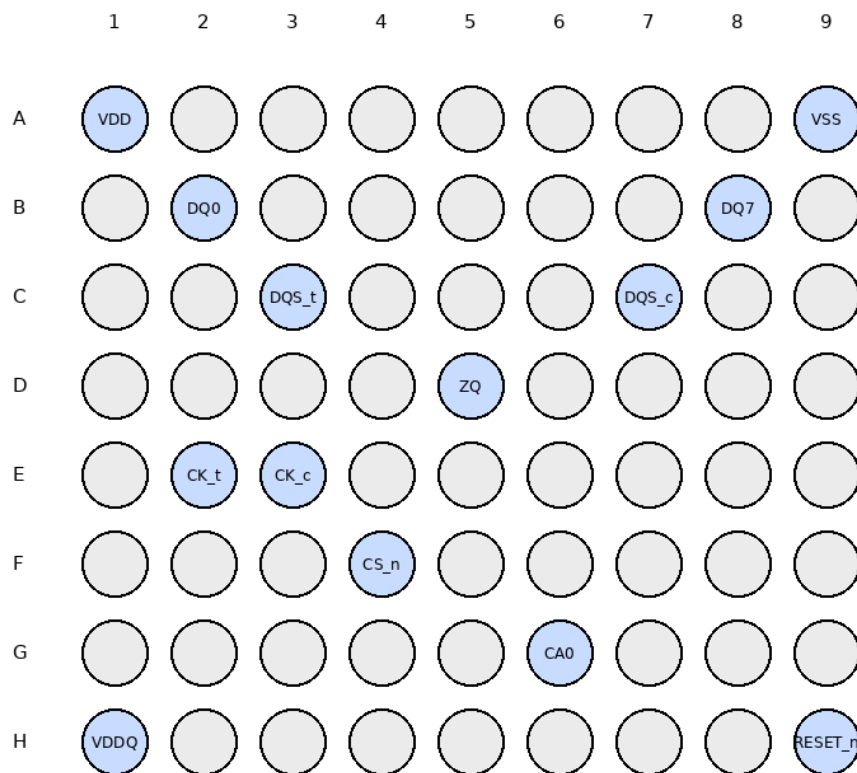


Figure 1 — Ball Assignment (Top View)

The ball pitch is 0.8 mm in both directions; package outline dimensions are defined in JESD-SYN-02.

5 Electrical Characteristics

5.1 DC Operating Conditions

All voltages are referenced to VSS. Operation outside the limits of Table 2 is not guaranteed.

Table 2 — Recommended DC Operating Conditions

Symbol	Parameter	Rating			Unit	Note
		Min	Typ	Max		
VDD	Core supply voltage	1.067	1.100	1.133	V	1
VDDQ	I/O supply voltage	0.475	0.500	0.525	V	1
VPP	Wordline supply voltage	1.710	1.800	1.890	V	2
IDD2N	Precharge standby current	-	42	55	mA	
TOPER	Operating case temperature	0	-	85	°C	3

NOTE 1 DC bandwidth is limited to 20 MHz.

NOTE 2 VPP must be equal to or greater than VDD at all times.

NOTE 3 Refresh rate doubles ($t_{REFI} = 1.95 \mu s$) above 85 °C.

5.2 AC Timing Parameters

Timing parameters are specified for all supported speed bins. Values in nCK are rounded up to the next integer clock.

Table 3 — Timing Parameters

Parameter	Symbol	DDR6-6400	DDR6-9600	Unit
Average clock period	$t_{CK(avg)}$	0.312	0.208	ns
ACT to internal read or write delay	t_{RCD}	14.375	14.000	ns
Row precharge time	t_{RP}	14.375	14.000	ns
ACT to PRE command period	t_{RAS}	32	32	ns
ACT to ACT or REF command period	t_{RC}	46.375	46.000	ns
Write recovery time	t_{WR}	30	30	ns
Refresh cycle time (16 Gb)	t_{RFC1}	295	295	ns
Average periodic refresh interval	t_{REFI}	3.9	3.9	μs

Table 3 — Timing Parameters (Cont'd)

Parameter	Symbol	DDR6-6400	DDR6-9600	Unit
CAS to CAS delay, same bank group	tCCD_L	8	12	nCK
CAS to CAS delay, different bank group	tCCD_S	8	8	nCK
Read latency	RL	22	32	nCK
DQS output access time from CK_t/CK_c	tDQSCK	-0.15	-0.10	tCK(avg)

6 Command and Timing

6.1 Read Operation

A read burst is initiated by a RD command. The first data is driven $RL = 22$ nCK after the RD command for DDR6-6400, and the burst length is BL16. The data strobe DQS_t toggles with a preamble of 2 nCK.

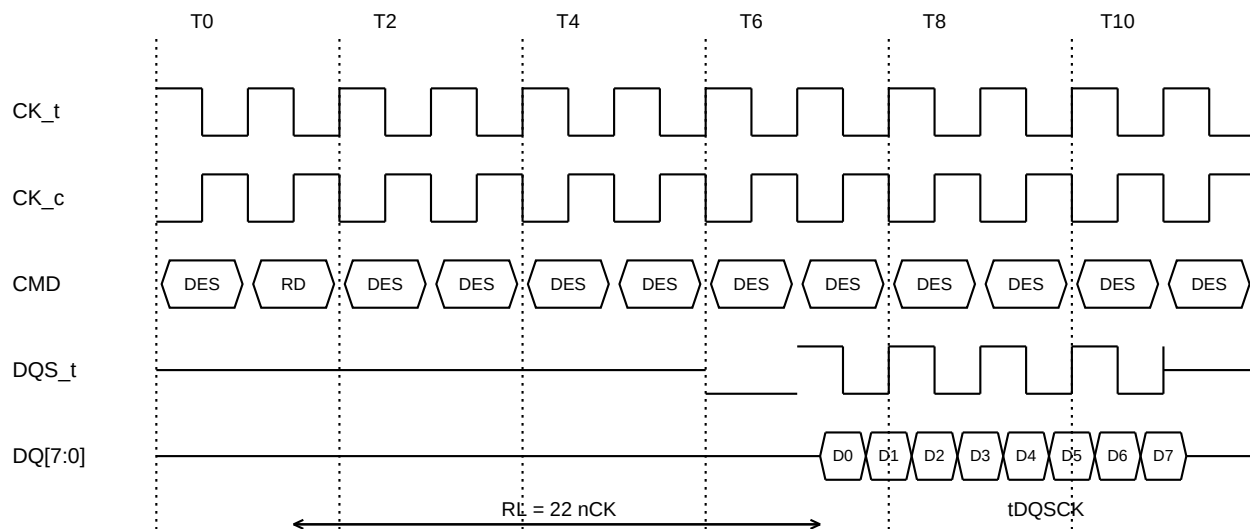


Figure 2 — Read Burst Timing (BL16, RL = 22)

NOTE 1 DES commands are shown for clarity; NOP is not supported.

6.2 Power-Down State Machine

The simplified state diagram in Figure 3 shows transitions between the IDLE, ACTIVE, PRECHARGE, POWER DOWN and SELF REFRESH states. Power-down entry (PDE) requires $t_{CKE} = 5 \text{ nCK}$.

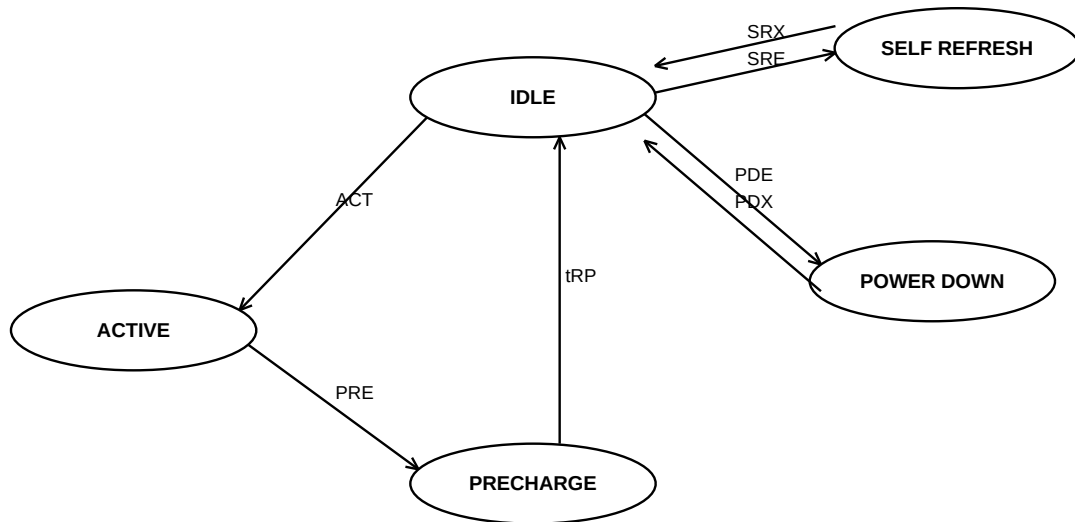


Figure 3 — Simplified Power-Down State Diagram

- Self refresh entry (SRE) is allowed only from IDLE.
- Exit from POWER DOWN (PDX) requires $t_{XP} = 7.5 \text{ ns}$.

Annex A (informative) Differences Between Revisions

This annex briefly describes the changes made to this synthetic standard.

A.1 Changes from Revision 0.9

- Added tDQSCK limits for DDR6-9600.
- Corrected VPP maximum to 1.890 V.